

Exam 1 Practice Problem Solutions

NOTE: For *many* of these problems, there are multiple possible mathematically equivalent correct answers.

1. $\langle 3, 2, 5 \rangle$
2. $\frac{1}{\sqrt{9}} \langle 2, 1, -2 \rangle$
3. Dot product is 0, cross product is $-2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$.
4. The first one tells you that they are orthogonal (at a ninety degree angle in their common plane) to each other. The second one tells you that they are parallel to one another.
5. 20
6. $-4\mathbf{j} + 2\mathbf{k}$
7. The line is entirely on the plane.
8. It does not.
9. $(1, 2, 1)$
10. $(x - 2) + (y - 3) - (z - 5) = 0$
11. $-8(x - 1) + 7(y - 1) + (z - 2) = 0$
12. $x - y = 1$
13. $x(t) = 1 + t, y(t) = 1 - 2t, z(t) = 2 - t$ (other, equivalent answers possible)
14. $x(t) = 1 + 5t, y(t) = -1 - t, z(t) = 1 - 3t$ (other, equivalent answers possible)
15. $x(t) = \cos(t), y(t) = \sin(t), \quad 0 \leq t \leq 2\pi$
16. $x(t) = 4\cos(t), y(t) = 4\sin(t), \quad 0 \leq t \leq 2\pi$
17. $x(t) = \cos(t), y(t) = 9\sin(t), \quad 0 \leq t \leq 2\pi$
18. $x(t) = 50\cos(t), y(t) = 30\sin(t), \quad 0 \leq t \leq 2\pi$
19.
$$(x(t), y(t)) = \begin{cases} (2t, 0), & 0 \leq t \leq 1 \\ (2 - t, 5(t - 1)), & 1 \leq t \leq 2 \\ (0, 5(3 - t)), & 2 \leq t \leq 3 \end{cases}$$
20. a) $(0, 2)$ and $(1, e + 1)$
b) $\langle 3t^2, e^t \rangle$
c) $\left\langle \frac{3t^2}{\sqrt{9t^4 + e^{2t}}}, \frac{e^t}{\sqrt{9t^4 + e^{2t}}} \right\rangle$
21. $f_x = 2xy \cos(x^2y), \quad f_y = x^2 \cos(x^2y)$

22. $f_x = \tan(x+y) + x \sec^2(x+y)$, $f_y = x \sec^2(x+y)$
23. $f_x = \frac{2x}{100 + x^2 + y^4}$, $f_y = \frac{4y^3}{100 + x^2 + y^4}$
24. $f_x = 3x^2y + 2xy^2 + 2$, $f_y = x^3 + 2x^2y - 1$, $f_{xx} = 6xy + 2y^2$, $f_{yy} = 2x^2$, $f_{xy} = 3x^2 + 4xy$
25. $f_x = 1 + 2xy^2 + 2xy^2z^2$, $f_y = 2x^2y + 2x^2yz^2$, $f_z = 2x^2y^2z$
26. $\frac{-2}{\sqrt{3}}$
27. Decreasing (directional derivative is $-3/\sqrt{2}$)
28. It is instantaneously constant.
29. a) $\langle ye^x, e^x \rangle$
 b) $\langle 0, 1 \rangle$
 c) $\frac{-2}{\sqrt{5}}$
 d) $\frac{-2}{\sqrt{5}}$
 e) 0
30. a) $\langle -2xy^2e^{-x^2y^2}, -2x^2ye^{-x^2y^2} \rangle$
 b) $\left\langle \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$
 c) $\frac{-2}{\sqrt{5}e}$
31. a) $\langle 2xy + y^2, x^2 + 2xy, 3z^2 \rangle$
 b) $\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$
 c) $12/\sqrt{6}$
32. $x + y + z = 3$
33. $2x + 2y + z = 4$
34. $2x - z = 2$
35. $x - y + 2z = 1$
36. It is not a critical point; $f_y \neq 0$.
37. Local minimum
38. Saddle point
39. (a) $(-2, -2)$ is the only critical point.
 (b) $(-2, -2)$ corresponds to a local maximum; the local maximum value of f at this point is $f(-2, -2) = 8$.

40. (a) $(0, 0)$ and $(2, 2)$ are two critical points.
 (b) Critical point $(0, 0)$ corresponds to a saddle point; critical point $(2, 2)$ corresponds to a local maximum. The local maximum value of f is $f(2, 2) = 8$.
41. $\frac{df}{dt} = (2xy + 3y^4)(2\cos(2t)) + (x^2 + 12xy^3)(-\sin(t))$
42. $\frac{df}{dt} = 2xe^{x^2-y}(4\cos(4t)) - e^{x^2-y}(3t^2)$
43. $\frac{df}{dt} = -y\cos(xy) - x\cos(xy)\frac{1}{t} + e^{t-1}$
44. $\frac{df}{dt} = (4x^3 - 4y^2)\left(\frac{1}{2}t^{-1/2}\right) - 16xye^{2t} - 3z^2t^{-2}$
45. $\frac{\partial z}{\partial s} = 4xys + 2s + 2x^2; \quad \frac{\partial z}{\partial t} = 6xt - x^2 + 3$
46. $\frac{\partial w}{\partial t} = y(2te^s) + x\cos(s); \quad \frac{\partial w}{\partial s} = y(t^2e^s) + x(-t\sin(s))$
47. $f(x, y) = y^2 - x^2, \quad f_x = -2x, \quad f_y = 2y.$

- (a) At $(1, 0)$, holding $y = 0$ fixed: $f(x, 0) = -x^2$, which is **decreasing** as x increases. This tells us $\frac{\partial f}{\partial x} < 0$ at $(1, 0)$.

$$f_x(1, 0) = -2(1) = -2 < 0. \checkmark$$

Graph confirmation: near $(1, 0)$, moving in the direction of increasing x , the surface becomes bluer (smaller z) — consistent with f decreasing.

- (b) At $(0, 1)$: $f_x(0, 1) = -2(0) = 0$, and $f_y(0, 1) = 2(1) = 2 > 0$. So $\frac{\partial f}{\partial y} > 0$ at this point.

Graph confirmation: near $(0, 1)$, moving in the direction of increasing y , the surface rises toward the red (larger- z) region — consistent with $f_y > 0$.

- (c) Critical point: solving $f_x = -2x = 0$ and $f_y = 2y = 0$ gives $(0, 0)$.

$$f_{xx} = -2, \quad f_{yy} = 2, \quad f_{xy} = 0, \quad D = f_{xx}f_{yy} - f_{xy}^2 = (-2)(2) - 0 = -4 < 0.$$

Since $D < 0$, $(0, 0)$ is a **saddle point**. This matches the graph: the surface rises along the y -axis and falls along the x -axis near the origin.

48. $f(x, y) = x^2 + y^2, \quad f_x = 2x, \quad f_y = 2y.$

- (a) At $(1, 0)$, holding $y = 0$ fixed: $f(x, 0) = x^2$, which is **increasing** as x increases. This tells us $\frac{\partial f}{\partial x} > 0$ at $(1, 0)$.

$$f_x(1, 0) = 2(1) = 2 > 0. \checkmark$$

Graph confirmation: near $(1, 0)$, moving in the direction of increasing x , the surface becomes redder (larger z) — consistent with f increasing.

- (b) At $(0, -1)$: $f_x(0, -1) = 2(0) = 0$, and $f_y(0, -1) = 2(-1) = -2 < 0$. So $\frac{\partial f}{\partial y} < 0$ at this point.

Graph confirmation: near $(0, -1)$, moving in the direction of increasing y (toward the origin), the surface descends toward the blue (smaller- z) minimum — consistent with $f_y < 0$.

- (c) Critical point: solving $f_x = 2x = 0$ and $f_y = 2y = 0$ gives $(0, 0)$.

$$f_{xx} = 2, \quad f_{yy} = 2, \quad f_{xy} = 0, \quad D = f_{xx}f_{yy} - f_{xy}^2 = (2)(2) - 0 = 4 > 0, \quad f_{xx} > 0.$$

Since $D > 0$ and $f_{xx} > 0$, $(0, 0)$ is a **local minimum**. This matches the graph: the surface is a bowl opening upward, with its lowest point at the origin.

49. $f(x, y) = 18 - x^2 - 3y^2$, $f_x = -2x$, $f_y = -6y$. Point: $(2, 1)$, with $\mathbf{v} = \langle 4, -3 \rangle$.

- (a)

$$\nabla f(2, 1) = \langle f_x(2, 1), f_y(2, 1) \rangle = \langle -2(2), -6(1) \rangle = \langle -4, -6 \rangle.$$

- (b) The gradient always points perpendicular to the level curve, in the direction in which f increases most rapidly. Since $f(2, 1) = 11$ and the neighboring curve $f = 13$ lies closer to the origin, $\nabla f(2, 1) = \langle -4, -6 \rangle$ should be drawn perpendicular to the $f = 11$ curve, pointing inward (toward the origin) — toward that higher-valued curve.

- (c) Direction: $\mathbf{v} = \langle 4, -3 \rangle$, $|\mathbf{v}| = 5$, so $\mathbf{u} = \langle \frac{4}{5}, -\frac{3}{5} \rangle$.

$$D_{\mathbf{u}}f(2, 1) = \nabla f(2, 1) \cdot \mathbf{u} = (-4) \left(\frac{4}{5} \right) + (-6) \left(-\frac{3}{5} \right) = \frac{-16+18}{5} = \frac{2}{5} > 0.$$

Since $D_{\mathbf{u}}f(2, 1) > 0$, f is **increasing** in the direction of $\mathbf{v}$.

Graph confirmation: the red arrow in Figure (b) points from the $f = 11$ curve toward the $f = 13$ curve — i.e., toward higher level-curve values — consistent with f increasing in that direction.

50. $f(x, y) = x^2 - y^2$, $f_x = 2x$, $f_y = -2y$. Point: $(1, 2)$, with $\mathbf{v} = \langle 3, 1 \rangle$.

- (a)

$$\nabla f(1, 2) = \langle f_x(1, 2), f_y(1, 2) \rangle = \langle 2(1), -2(2) \rangle = \langle 2, -4 \rangle.$$

- (b) The gradient always points perpendicular to the level curve, in the direction in which f increases most rapidly. Since $f(1, 2) = -3$ and the neighboring curve $f = -1$ lies closer to the x -axis (toward larger, less negative values), $\nabla f(1, 2) = \langle 2, -4 \rangle$ should be drawn perpendicular to the $f = -3$ curve, pointing toward that higher-valued curve.

- (c) Direction: $\mathbf{v} = \langle 3, 1 \rangle$, $|\mathbf{v}| = \sqrt{10}$, so $\mathbf{u} = \left\langle \frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right\rangle$.

$$D_{\mathbf{u}}f(1, 2) = \nabla f(1, 2) \cdot \mathbf{u} = (2) \left(\frac{3}{\sqrt{10}} \right) + (-4) \left(\frac{1}{\sqrt{10}} \right) = \frac{6-4}{\sqrt{10}} = \frac{2}{\sqrt{10}} > 0.$$

Since $D_{\mathbf{u}}f(1, 2) > 0$, f is **increasing** in the direction of $\mathbf{v}$.

Graph confirmation: the red arrow in Figure (b) points from the $f = -3$ curve toward the $f = -1$ curve — i.e., toward higher (less negative) level-curve values — consistent with f increasing in that direction.